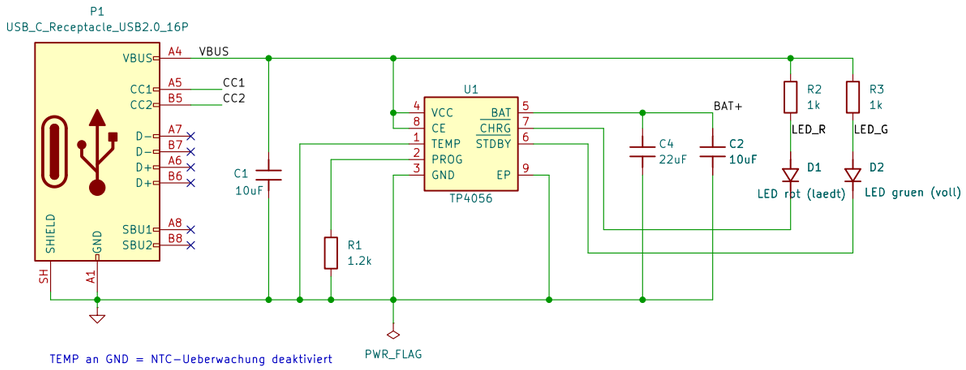
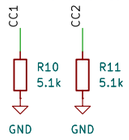
Akku 18650 -> 3,3V stabil (2,4 ... 4,2V Eingang)

1 Laden: USB 5V -> TP4056 (ca. 0,9A mit R1 = 1,2k)



TEMP an GND = NTC-Ueberwachung deaktiviert



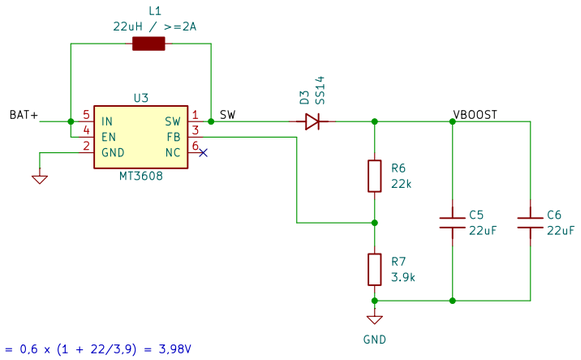
USB-C: CC1/CC2 je 5,1k nach GND (Sink -> 5V)

2 Akku 18650 + Schutz (DW01A / FS8205A)

Zelle zwischen BAT+ und BAT-. Systemmasse GND = P- (hinter FS8205A).

Abschaltung: Ueberladung 4,3V / Tiefentladung 2,4V / Ueberstrom

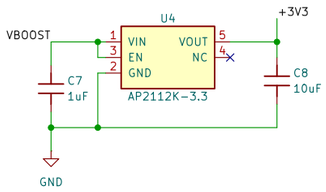
3 Boost MT3608 -> 4,0V

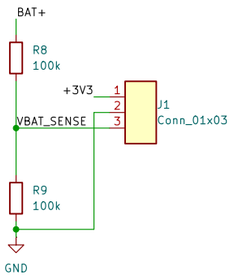


Vout = 0,6V x (1 + R6/R7) = 0,6 x (1 + 22/3,9) = 3,98V

Fuer 5V: R7 = 3k. Bei vollem Akku (4,2V) taktet der Boost nur noch wenig, LDO regelt.

4 LDO AP2112K-3.3 -> 3,3V stabil





VBAT_SENSE = Akkuspannung / 2 fuer ADC (bei ~1,5V = 3,0V Akku -> Schlafmodus)

TP4056 / DW01A+FS8205A / MT3608 / AP2112K-3.3

Sheet: /
File: Akku_3V3_Versorgung.kicad_sch

Title: Akku 18650 -> 3,3V stabil

Size: A3	Date: 2026-10-10	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1